INVESTIGATION OF ZIRCONIUM AND GRAPHITE EFFECTS ON LM SERIES ALUMINIUM ALLOY HYBRID COMPOSITES PRODUCED VIA STIR CASTING

A PROJECT REPORT

Submitted by

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in partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING
IN
MECHANICAL ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
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MAY 2026

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Under the Guidance of

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BONAFIDE CERTIFICATE

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We declare that to the best of our knowledge the work reported here in has been composed solely by ourself and that it has not been in whole or in part in any previous application for a degree.

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ACKNOWLEDGEMENT

We thank the almighty god without his blessing it would not have been possible for us to complete this project.

At this moment of having successfully completed our project, we wish to convey our sincere thanks and gratitude to our management of our college and our beloved chairman **Dr. K. RAMAKRISHNAN, B.E., Ph.D., who** provide all the facilities to us.

We would like to express our sincere thanks to our Executive Director

Dr**. S. KUPPUSAMY, M.B.A., Ph.D**., for forwarding us to do our project and offering adequate duration in completing our project.

We are also grateful to our Principal **Dr. D. SRINIVASAN, Ph.D.,** for our constructive suggestion and encouragement during our project.

We wish to express our profound thanks to **Dr. H. RAMAKRISHNAN, Ph.D**., Head of the Mechanical Engineering Department, for providing all necessary facilities for doing this project.

We wholeheartedly and sincerely acknowledgement our deep sense of gratitude and indebtedness to our beloved guide **Mr.B.PRAKASH,M.Tech.,(ph.D.),** Associate professor, Mechanical Engineering Department, Tiruchirappalli, For his expert guidance and encouragement throughout the duration of the project.

We extend my gratitude to all the staff members of Mechanical Engineering Department, **K.RAMAKRISHNAN COLLEGE OF ENGINEERING (AUTONOMOUS)** for their kind help and valuable support to complete the project successfully. We would like to thank my

parents, and friends who have always been a constant source of support in our project.

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PROGRAM OUTCOMES (PO's)	
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PO 6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice
PO 7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
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PO 10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO 12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

Aluminium metal matrix composites (AMCs) are widely used in aerospace, automotive, and structural applications due to their superior strength-to-weight ratio and enhanced mechanical properties. This study focuses on the investigation of zirconium (Zr) and graphite (Gr) effects on LM series aluminium alloy hybrid composites fabricated through the stir casting process. Zirconium is introduced to improve grain refinement and mechanical strength, while graphite acts as a solid lubricant to enhance wear resistance. The composites are prepared by varying the weight percentages of zirconium and graphite in the LM aluminium alloy matrix. Stir casting ensures uniform distribution of reinforcements under controlled temperature and stirring conditions. Mechanical properties such as tensile strength, hardness, and wear resistance are evaluated, along with microstructural analysis using optical microscopy. The results indicate that the addition of zirconium significantly enhances strength and hardness due to grain refinement, while graphite improves tribological properties by reducing friction and wear. The hybrid combination demonstrates a balance between strength and self-lubricating behavior, making it suitable for advanced engineering applications. The experimental results indicate that zirconium significantly improves the mechanical strength and hardness due to grain refinement, whereas graphite enhances tribological performance by reducing friction and wear. The hybrid combination of zirconium and graphite provides a balanced improvement in both mechanical and wear properties, making the composite suitable for automotive and application.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Metal matrix composites (MMCs) are widely used in aerospace, automotive, and structural applications due to their superior strength-to-weight ratio, improved wear resistance, and enhanced mechanical properties. Among various matrix materials, zirconium alloys are known for their excellent corrosion resistance, high temperature stability, and good mechanical strength, making them suitable for advanced engineering applications.

This project focuses on the fabrication and investigation of zirconium alloy reinforced with LM series aluminium alloy and graphite, aiming to improve its mechanical and tribological properties. The LM series aluminium alloy is introduced to provide lightweight characteristics and improve overall strength, while graphite acts as a solid lubricant, enhancing wear resistance and reducing friction during operation.

LM Series-based hybrid metal matrix composites have gained attention due to their ability to combine high strength with excellent thermal and chemical stability. The incorporation of Zirconium Alloy and graphite into the zirconium matrix is expected to produce a composite with improved load-bearing capacity, reduced wear, and enhanced thermal performance.



Fig. LM Series Aluminium Alloy

LM series aluminium alloys are used as the base matrix material ,widely used cast alloys known for their lightweight nature, excellent castability, and good corrosion resistance. In this composite, LM aluminium alloy contributes to reducing the overall weight while maintaining adequate strength. It also improves the fluidity during the casting process, ensuring better distribution of reinforcements and enhanced bonding with in the composite.

Zirconium alloy used as the reinforcement material in this study. It provides high strength, excellent corrosion resistance, and thermal stability. Due to its ability to form a protective oxide layer, zirconium exhibits superior resistance to chemical attack and performs well under extreme conditions. These properties make it highly suitable for structural and high-temperature applications.This project focuses on the fabrication and investigation of zirconium alloy reinforced with LM series aluminium alloy and graphite, aiming to improve its mechanical and tribological properties.



Fig.Graphite

Graphite is used as a reinforcement material due to its unique layered structure and solid lubricating properties. It significantly reduces friction and wear during sliding conditions, thereby improving the tribological performance of the composite. Additionally, graphite enhances thermal conductivity and helps in dissipating heat effectively.

Graphite improves wear resistance, reduces friction, and minimizes heat, while LM aluminium alloy provides lightweight strength, good castability, and corrosion resistance.

The combination of zirconium alloy as the matrix and LM series aluminium alloy along with graphite as reinforcements results in a hybrid composite with improved strength, reduced wear, enhanced thermal performance, and better overall efficiency. Such composites are highly suitable for applications in aerospace, automotive, and other high-performance engineering fields where durability, lightweight, and reliability are essential.

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Fig.Zirconium

The present investigation on LM series aluminium alloy hybrid composites reinforced with zirconium and graphite has demonstrated significant improvements in both mechanical and tribological properties. The stir casting process proved to be an effective and economical method for fabricating the composites with fairly uniform distribution of reinforcements.

1.1 Composition Material

The hybrid composite material under investigation consists of a zirconium alloy matrix reinforced with LM series aluminium alloy and graphite particles. The choice of zirconium alloy as the matrix material is due to its excellent corrosion resistance, high temperature stability, and good mechanical strength. The addition of LM series aluminium alloy and graphite as reinforcements enhances the overall performance of the composite by improving strength, wear resistance, and thermal properties, making it suitable for advanced engineering applications.

base Matrix:

- LM Series Aluminium Alloy: A high-performance metallic material known for its excellent corrosion resistance, thermal stability, and mechanical strength. It forms a protective oxide layer, ensuring durability in harsh environments and making it suitable for structural and high-temperature applications.

Reinforcements:

1. Zirconium :

    - Type: Elemental or intermetallic lithium particles.
    - Function: Zirconium acts as a grain refiner in LM aluminium composites by forming fine Al_3Zr particles, which produce a uniform microstructure and improve strength and hardness. It also enhances thermal stability and bonding between the matrix and graphite, reducing defects and improving overall composite performance.
    - Weight %: Typically 0.5 wt% – 2 wt% (most common range)

2. Graphite Nanoplatelets (GNPs):
    - Type: Solid lubricant (carbon-based layered structure).
    - Function: Graphite improves wear resistance and reduces friction due to its self-lubricating nature. It also enhances thermal conductivity and reduces heat generation during operation.
    - Weight %: Generally 1–3 wt% for uniform dispersion and optimal tribological performance

Hybrid Composite Composition (Example):

MATERIAL	WEIGHT %
LM series	96.0
Zirconium	2.0
Graphite nanoplatelets	2.0

Table 1.1 Composite Composition

Fabrication Consideration:

Stir casting or powder metallurgy is typically used to ensure homogeneous distribution of reinforcements. Proper surface treatment and dispersion techniques are crucial to prevent agglomeration of graphene and to ensure interfacial bonding with the aluminum matrix. During stir casting, proper control of stirring speed and time is essential to ensure homogeneous mixing of the reinforcements in the molten aluminium. Temperature control is also critical, as LM25 must be melted at an appropriate temperature (around 700–750°C) to avoid oxidation and ensure proper mixing.

1.2 Metal composition

1. Base Alloy –LM Series of Aluminium Alloy:

LM25 aluminium alloy is an aluminium-silicon casting alloy containing silicon as the major alloying element in the range of 6.5–7.5%, which improves fluidity and casting characteristics. Magnesium is present in small amounts (0.2–0.6%) to enhance strength through heat treatment. The remaining composition consists mainly of aluminium, along with trace amounts of iron, copper, manganese, nickel, zinc, lead, and tin. This combination provides LM25 with good corrosion resistance, moderate strength, and excellent castability, making it suitable as a base matrix material in metal matrix composites.

ELEMENT	TYPICAL WEIGHT %
Aluminium (Al)	Balance(~85%-90%)
Silicon (Si)	6.5 -- 7.5
Magnesium (Mg)	0.3 – 0.9
Iron (Fe)	0.05 – 0.3
Copper (Cu)	0.1 – 0.4
Manganese (Mn)	0.1 – 0.3
Nickel(Ni0	0.05 – 0.2
Zinc (Zn)	0.01 – 0.1
Others (each)	< 0.05

Table 1.2 Metal composition

2. Reinforcements:

a. Zirconium– Metal Reinforcement

• Purpose: Strength enhancement, grain refinement, improves hardness, and increases thermal stability.
• Typical Addition Level: 1 – 5 wt%
• Effect on Matrix:

- Forms fine intermetallic compounds such as Al_3Zr.
- Refines grain structure and improves mechanical properties.
- Improves fatigue and corrosion resistance.

b. Graphite – Nano-reinforcement

- Purpose: Reduces friction, improves wear resistance, provides self-lubrication, and enhances machinability.
- Typical Addition Level: 0.5 – 3 wt%
- Effect on Matrix:

  - Acts as a solid lubricant and reduces coefficient of friction.
  - Improves wear resistance and surface finish.
  - Reduces heat generation during sliding conditions..

1.3 Metal Reinforcement:

Metal reinforcements are incorporated into aluminum matrix composites (AMCs) to improve mechanical, thermal, and wear properties. In the context of LM Series-based hybrid composites, zirconium is used as a metallic reinforcement due to its unique properties and synergistic interaction with the aluminum matrix.

1.3.1 Role of Zirconium as a Metal Reinforcement

Zirconium is an effective metallic reinforcement that plays a significant role in improving the microstructure and mechanical performance of aluminium matrix composites:

- Grain Refinement**:** Zirconium forms fine intermetallic compounds such as Al_3Zr, which act as nucleation sites and refine the grain structure, leading to improved mechanical properties.
- Strength Enhancement**:** The presence of zirconium increases the strength and hardness of the composite by hindering dislocation movement within the matrix.
- Thermal Stability: Zirconium improves high-temperature performance stabilizing the microstructure and preventing grain growth during thermal exposure.
- Wear Resistance**:** The addition of zirconium enhances resistance to wear and deformation, making the material suitable for applications involving friction and load

1.3.2 Interaction with Graphite

In a hybrid composite system, zirconium works synergistically with graphite reinforcement to provide a balanced improvement in both mechanical and tribological properties:

- Zirconium enhances the bonding between the LM series aluminium matrix.
- It helps in stress transfer between the graphite and LM series matrix due to better wettability and bonding characteristics.
- Thermal Properties: Lithium-modified aluminum alloys exhibit improved thermal conductivity and reduced thermal expansion, aiding in dimensional stability under thermal cycling

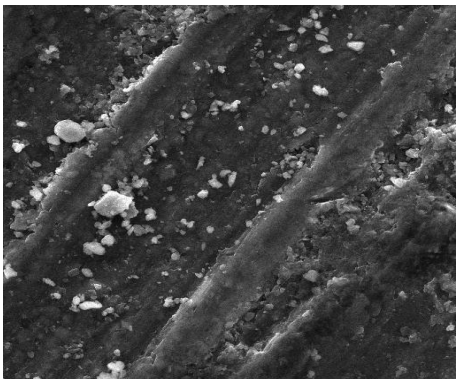


Fig 1.4 ZIRCONIUM REINFORCE IN IN LM SERIES AND GRAPHITE15%



Fig 1.5 ZIRCONIUM REINFORCE IN LM SERIES AND GRAPHITE20%

1.3.3 Selection Justification

The choice of lithium as a reinforcement in LM series alloy is driven by:

- Its ability to reduce weight while maintaining or enhancing strength.
- Its compatibility with aluminum and ability to form fine-grained microstructures.
- Its enhancement of corrosion resistance in controlled environments.

1.4 Metal Factors Contributing to Mechanical Properties:

The mechanical behavior of LM series-based hybrid composites reinforced with lithium and graphene is influenced by the intrinsic properties and interactions of the constituent materials. The following subsections discuss how each metal or reinforcement contributes to the overall mechanical performance of the composite.

1.4.1 LM series aluminium alloy

LM series is a wrought aluminum alloy primarily composed of aluminum with minor alloying elements like iron, silicon, manganese, and others. Its contribution includes:

- High Strength-to-Weight Ratio: LM series offers a good balance between mechanical strength and lightweight characteristics, ideal for aerospace and automotive applications.
- Corrosion Resistance: The alloy naturally forms a passive oxide layer, improving durability.
- Matrix Phase: Acts as the continuous phase in the composite, binding the reinforcement (lithium and graphene) and transferring load efficiently.

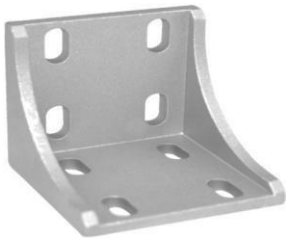


Fig 1.6 LM series

1.4.2 Zirconium

Zirconium, when added to aluminum alloys, enhances mechanical properties and thermal stability. It forms fine intermetallic compounds (such as Al_3Zr), which improve the overall performance of the alloy. Its contributions are:

- Grain Refinement: Zirconium promotes the formation of fine, stable grains, improving strength and toughness.
- Increased Stiffness: Lithium increases Young's modulus of the aluminum matrix.
- Enhanced Strength: The formation of Al_3Zr precipitates increases yield strength and creep resistance.



Fig 1.7 Zirconium metal

1.4.3 Graphite

Graphite, a form of carbon, is widely used as a reinforcement material in aluminum matrix composites due to its excellent lubricating and thermal properties. Its contributions are:

- Ultra-High Strength: Graphite has a tensile strength of ~130 GPa, significantly enhancing the composite's overall strength.
- Load Transfer Efficiency: Due to its large surface area and strong interfacial bonding, graphene aids in effective stress distribution.
- Crack Resistance: Acts as a barrier to crack propagation, improving toughness and wear resistance.



Fig 1.8 Graphite

1.5 Metal Matrix:

In this study, LM series aluminium alloy is used as the base matrix material. To enhance its mechanical and physical properties, a hybrid reinforcement of zirconium and graphite is incorporated at varying weight percentages: 5%, 10%, and 15%. These percentages represent the total weight of the reinforcement materials added to the metal matrix.

Composition Details:

Composite Sample	LM Series (%)	Zirconium (%)	Graphite (%)	Total Reinforcement (%)
Sample A	95	2.5	2.5	5
Sample B	90	5.0	5.0	10
Sample C	85	7.5	7.5	15

Table 1.3 Composition Details

Purpose of Varying Reinforcement Levels:

- 5% Reinforcement: Intended to observe the initial improvement in mechanical strength, hardness, and thermal behavior.
- 10% Reinforcement: Aimed at achieving optimal balance between strength enhancement and ductility.
- 15% Reinforcement: Used to study the upper limit of reinforcement without compromising structural integrity due to potential agglomeration or poor bonding.

1.6 Metal Polymer Matrix Material:

In advanced composite materials, matrix selection plays a critical role in determining overall performance, especially in hybrid systems where synergistic reinforcement is expected. In this study, the matrix material is an Aluminium series, which is further reinforced with zirconium and graphite, forming a metal- polymer hybrid composite.

AL Series Aluminium Alloy as the Base Metal Matrix:

Al 8085 alloy is a high-strength aluminium alloy typically used in structural applications due to its:

- Excellent strength-to-weight ratio
- Good thermal conductivity
- Corrosion resistance
- Light weight

This alloy is particularly suitable for aerospace and automotive industries where lightweight and high-performance materials are essential.

Hybrid Reinforcement: Zirconium And Graphite:

To enhance the mechanical, thermal, and wear properties of LM series aluminium alloy, zirconium and graphite are introduced as reinforcements:

- Zirconium, when added, refines the grain structure and improves thermal stability, thereby increasing strength and resistance to high-temperature deformation.
- Graphite, being a solid lubricant, improves wear resistance and reduces friction, thereby enhancing the tribological performance of the composite.

1.7 Advantages and Disadvantages of Composition:

1.7.1 Advantages:

- Improved Mechanical Properties
- Weight Reduction
- Reduced Friction due to Graphite Lubrication
- Enhanced Wear Resistance
- Good Corrosion Resistance
- Tailorable Properties based on reinforcement percentage

1.7.2 Disadvantages:

- Agglomeration of Reinforcements (Zirconium and Graphite)
- Difficulty in Uniform Distribution during Stir Casting
- Possibility of Porosity Formation
- Limited Ductility at Higher Reinforcement Content
- Interfacial Bonding Issues

CHAPTER 2

Literature Review

2.1 Introduction

Zirconium alloy have gained significant attention in recent years due to their superior mechanical and thermal properties compared to conventional materials. Among various aluminum alloys, zirconium alloy stands out for its excellent strength-to-weight ratio, corrosion resistance, and thermal stability, making it suitable for aerospace and automotive applications.

The incorporation of reinforcements such as graphite and LM series into zirconium alloy has emerged as a promising method to enhance their physical and mechanical properties. Graphite, known for its exceptional strength, conductivity, and surface area, acts as a nano-reinforcement that improves hardness, tensile strength, and thermal conductivity. LM series, when added in small amounts, contributes to further weight reduction and improved specific strength due to its low density and reactivity.

Hybrid composites that utilize both lithium and graphene reinforcements within the zirconium alloy have the potential to synergistically combine the advantages of both reinforcements. However, to fully exploit these materials, detailed investigations are needed concerning their fabrication methods, microstructural evolution, and performance under mechanical and thermal stresses.

2.2 Inferences from the Literature:

- Zirconium-Reinforced Aluminium Composites: Studies indicate that the addition of zirconium to aluminium alloys promotes grain refinement, resulting in improved tensile strength, hardness, and better structural stability. Zirconium also enhances interfacial bonding between matrix and reinforcement.

- Graphite-Reinforced Aluminium Composites: Research shows that graphite addition significantly improves wear resistance and reduces friction due to its solid lubricating nature. However, higher graphite content may lead to a reduction in hardness. Hybrid Reinforcements: Recent research indicates that combining multiple reinforcements (e.g., ceramic particles with graphene or metal additives like lithium) can result in synergistic property improvements. These include enhanced strength-to-weight ratio, improved thermal stability, and better fatigue resistance.
- Hybrid Reinforcements: Recent investigations reveal that combining ceramic or metallic reinforcements (such as zirconium) with solid lubricants (like graphite) produces a synergistic effect, improving both mechanical and tribological properties simultaneously.
- Fabrication Techniques: Stir casting is widely used due to its simplicity and cost-effectiveness in producing aluminium matrix composites, though achieving uniform distribution of reinforcements remains a challenge.
- Microstructural Characteristics: Uniform dispersion of reinforcements leads to enhanced load transfer and improved performance, while agglomeration and porosity negatively affect composite properties.

2.3 Problem Statement:

While extensive research has been carried out on aluminium-based composites reinforced with individual materials such as ceramic particles or solid lubricants, there exists a research gap in understanding the combined effects of zirconium and graphite in LM series aluminium alloy matrices. The development of hybrid composites using these reinforcements through stir casting presents several challenges that need to be addressed. The challenges lie in:

- Achieving uniform distribution of zirconium and graphite particles without agglomeration.
- Managing interfacial reactions and microstructural stability during processing.

CHAPTER 3

Material and Methodology

3.1 Experimental Methodology

The study involves the synthesis and characterization of Zirconium alloy-based hybrid metal matrix composites reinforced with LM Series and graphite. The methodology consists of the following steps:

1. Material Selection: Zirconium alloy is selected as the matrix due to its excellent mechanical properties and corrosion resistance. Lithium and graphene are selected as reinforcements for their lightweight and superior mechanical/thermal properties.
2. Powder Preparation: LM Series and graphite powders are weighed according to the desired weight percentages (typically 1–5 wt%) and thoroughly mixed with Al 8085 powder using ball milling to ensure uniform distribution.
3. Composite Fabrication: The hybrid composites are fabricated using powder metallurgy or stir casting techniques. In stir casting, the molten Al 8085 is stirred at high speed while reinforcements are added gradually to ensure proper dispersion.
4. Compaction and Sintering: In powder metallurgy, the mixed powders are compacted using a hydraulic press followed by sintering in a controlled atmosphere furnace.
5. Heat Treatment: Post-sintering or casting, heat treatment such as solutionizing and artificial aging is performed to enhance mechanical properties.
6. Testing and Characterization:

    - Mechanical Testing: Hardness (Brinell or Vickers), tensile strength, impact strength.
    - Microstructural Analysis: SEM and EDX to study the dispersion of reinforcement

- Wear Testing: Pin-on-disc wear tests to evaluate tribological behavior.
- Thermal Analysis: DSC/TGA to assess thermal stability.

3.2 Benefits of Adding LM Series Aluminium Alloy and Graphite

- Graphite:
    - Reduces friction due to its solid lubricating property.
    - Improves wear resistance of the material
    - Minimizes heat generation during sliding
- LM Series Aluminium Alloy:
    - Provides lightweight structure with good strength
    - Offers excellent castability for easy fabrication
    - Ensures good corrosion resistance for durability

The combination of aluminium and graphite in zirconium matrix creates a synergistic effect, resulting in a lightweight, high-strength, and wear-resistant composite suitable for aerospace and automotive applications.

3.3 Material Properties:

3.3.1 Zirconium Alloy*:*

- Density: ~6.5 g/cm³ (lower than many structural metals like steel)
- Melting Point: ~1855 °C
- Color: Silver-gray metal
- Thermal Conductivity: Moderate (~22 W/m·K)
- Coefficient of Thermal Expansion: L

3.3.1 LM Series:

- Density: 2.6-2.8 g/cm³
  Melting Point: ~660 °C
- Color: ~White
- Notable Features: Lightweight, excellent castability, good corrosion resistance, and strong compatibility with reinforcements for composite applications.

3.3.2 Graphite Properties:

- Density: ~2.2 g/cm³ (very lightweight)
- Structure: Layered (hexagonal) structure of carbon atoms
- Low Hardness: Soft material (acts as a lubricant)
- High Temperature Resistance: Stable up to ~3000°C in inert conditions
- Good Strength in Layers: Strong covalent bonding within layers
- Good Electrical Conductivity: Due to free electrons in carbon layers

3.4 Material Selection of Composite Laminate

The composite laminate is developed using as zirconium alloy the matrix material, with LM Series and graphite as reinforcements

- Zirconium Alloy: Known for high strength-to-weight ratio, corrosion resistance, and good workability. It is suitable for aerospace and structural applications.
- LM Series: Added in small amounts, lithium reduces the density of aluminum and increases its modulus, enhancing stiffness and specific strength.
- Graphite: A nano-scale carbon-based material known for exceptional mechanical, thermal, and electrical properties. It enhances the composite's tensile strength, hardness, and wear resistance.

3.5 Fabrication Process:

The stir casting method was employed for fabricating the Zirconium Alloy-LM Series-Graphite hybrid composite.

3.5.1 Stir Casting:

Stir casting is a liquid metallurgy technique where reinforcement particles are mechanically stirred into the molten metal to form a composite.

Procedure:

- Zirconium alloy is melted in an electric furnace at ~1840–1850°C.
- LM Series and graphite are preheated (~300–400°C) to remove moisture and improve wettability.
- The molten aluminum is stirred using a mechanical stirrer to create a vortex.
- Preheated lithium and graphene are gradually added into the vortex.
- Continuous stirring is maintained (typically 300–600 rpm for 5–10 minutes).
- The mixture is then poured into preheated moulds to form the desired laminate shape.

Why Stir Casting is Used for Zirconium Alloy, LM Series, and Graphite Composite:

- **Cost-Effective**: Among the cheapest methods for metal matrix composite (MMC) fabrication.
- **Simple and Scalable**: Easy to scale up for industrial applications.

- **Uniform Distribution**: Promotes relatively uniform dispersion of lithium and graphene in the aluminum matrix
- **Good Wettability**: Enhances the bonding between the matrix and reinforcements, especially when preheated and stirred properly.

3.5.3 Diagram of Stir Casting:

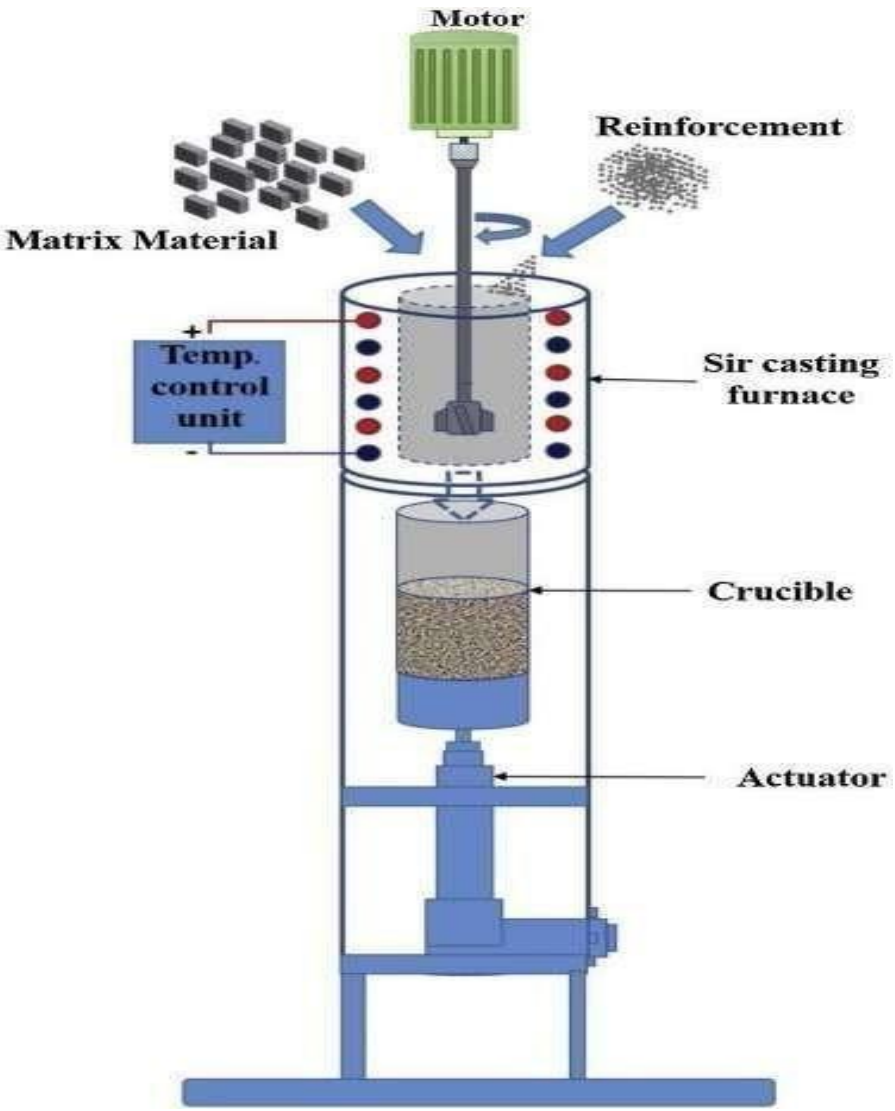


Fig 3.1 Stir casting

3.5 Flow Chart:

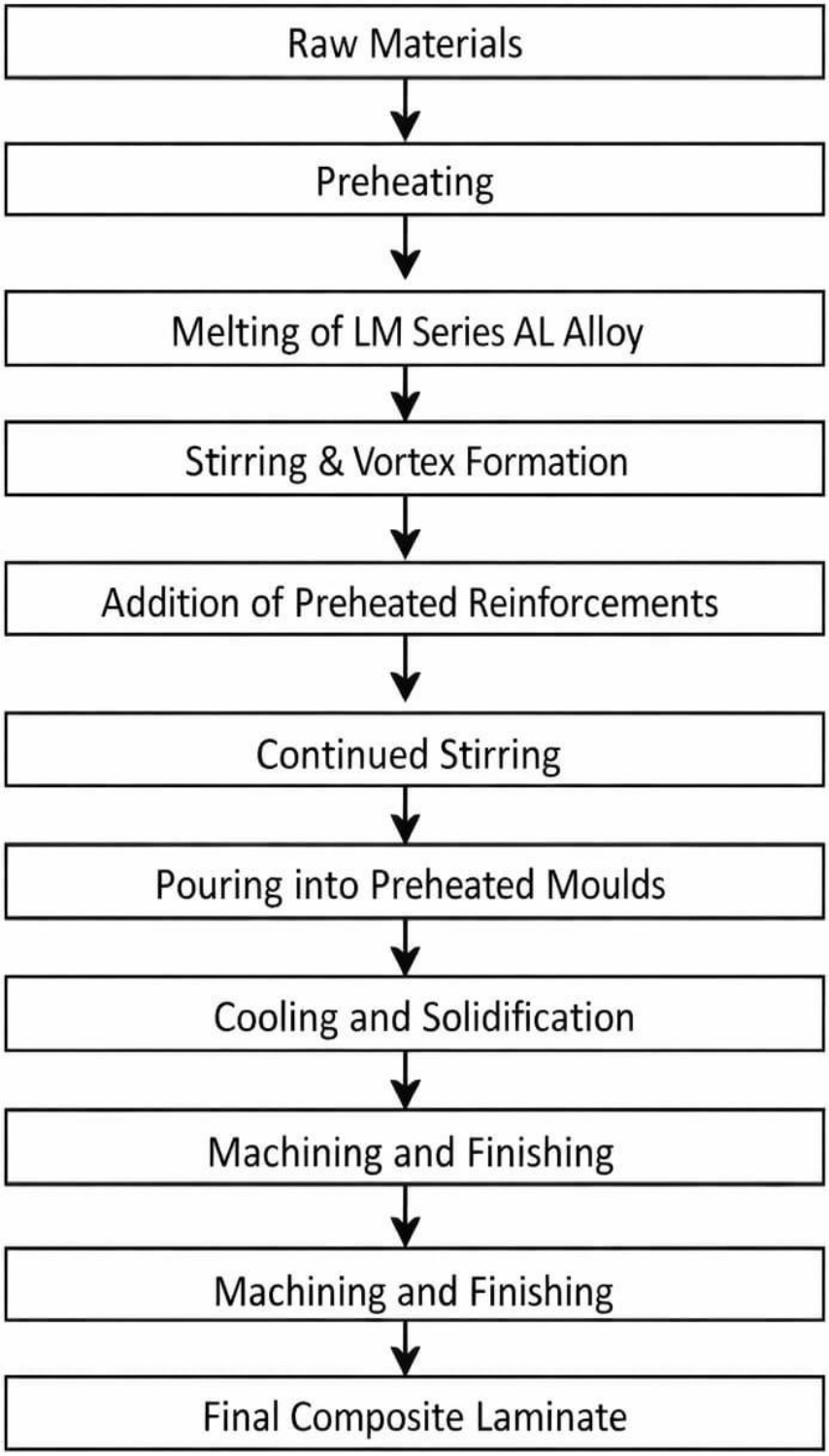


Fig 3.2 Experimental methodology

3.6 Cost Estimation Table:

S. No.	Item/Process	Details	Quantity (approx.)	Unit Cost (INR)	Total Cost (INR)
1.	LM series aluminium alloy	Base matrix metal	1kg	400/kg	400
2.	Zirconium	High-purity (3 wt%)	30g	15,000/kg	450
3.	Graphite nanoplatelets	Reinforcement (2 wt%)	20g	50,000/kg	1000
4.	Stir casting (or powder metallurgy)	Processing and mixing	Per kg basis	150/kg	150
5.	Pre-processing	Preheating, mixing, safety handling	Per kg basis	50/kg	50
6.	Post-processing	Machining, surface finishing	Per kg basis	100/kg	100
7.	Testing and characterization	SEM, XRD, tensile, hardness, corrosion tests	Per kg basis (avg.)	500/kg	500
	Total Estimated Cost (INR)	**(Per kg of composite)**			**2,650 /-**

Table 3.1 Cost Estimation

CHAPTER 4

Testing Methods and Result Discussion

4.1 Testing Methods:

4.1.1 Brinell Hardness Test:

Brinell Hardness Testing was performed according to ASTM E10 standards. A 10 mm diameter steel ball indenter was applied under a standard load of 500 kg-f for 15 seconds. The hardness values were calculated by measuring the diameter of the indentation. This test helped evaluate the resistance of the composite material to permanent surface deformation due to reinforcement effects.



Fig 4.2 Brinell Hardness Test Machine

4.1.2 Tensile Test:

Tensile properties were evaluated using a universal testing machine (UTM) according to ASTM E8/E8M standards. The test measured yield strength, ultimate tensile strength (UTS), and elongation at break. The reinforcement of graphene and lithium was expected to improve tensile properties due to their high strength and load transfer efficiency.



Fig 4.4 Tensile Test Machine

4.1.3 Impact Test:

Charpy impact tests were performed according to ASTM E23 to determine the toughness of the composite. Notched specimens were used to quantify the material's ability to absorb energy during sudden loading or fracture. This test provided insights into the ductility and energy-absorbing capacity of the composite under impact conditions.



Fig 4.5 Impact Test Machine

4.1.4 Test Results:

- SEM Analysis revealed a fairly uniform distribution of reinforcements with minimal agglomeration.

b) AL 8085 IN LITHIUM A N D GRAPHITE5%

b) ZIRCONIUM AND GRAPHITE15%

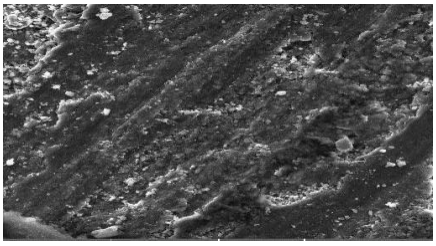




Fig 4.6 Sem Test Result

- Hardness values showed a significant improvement (approx. 15–25%) compared to base Al 8085 alloy, indicating effective dispersion and load-sharing by the reinforcements.

EQUIPMENT USED : METALLURGICAL MICROSCOPE- METSCOPE-1A				
LM Series Reinforce In Zirconium And Graphite	**HVI KG**	**HVI KG**	**HVI KG**	**AVG**
Sample 0%	59.8	60.4	60.8	60.4
Sample 5%	62.5	67.7	70.1	**67.7**
Sample 10%	70.7	74.7	77.9	**74.7**
Sample 15%	67.9	86.4	92.2	**86.4**

Table 4.1 Brinell Hardness Test Result

- Tensile Strength increased by 10–18%, while a slight reduction in elongation was noted, indicating a shift towards brittleness due to reinforcement particles.

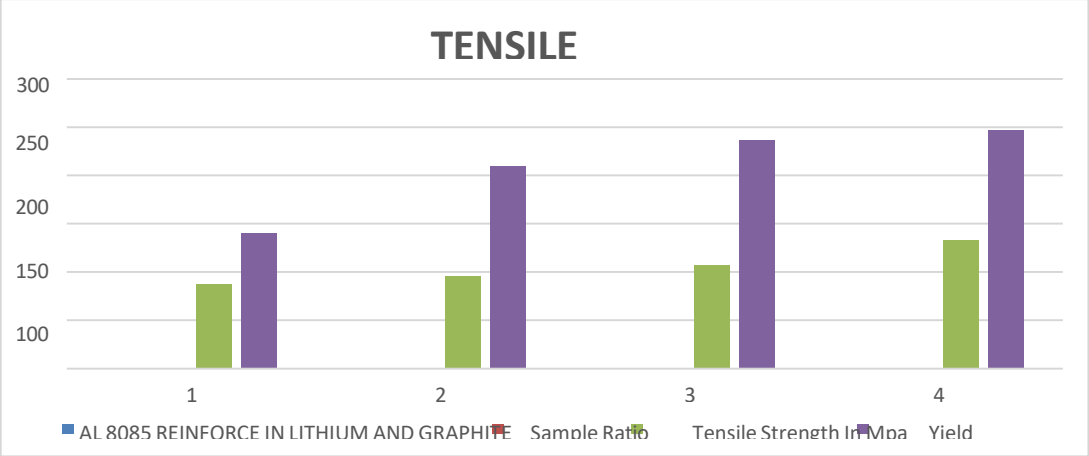


Fig 4.7 Tensile test

- Impact Strength showed moderate improvement, suggesting a balance between improved strength and ductility retention due to effective stress distribution.

Sample	**Area(mm^2)**	**Impact values(J)**	**Impact strength(J/mm^2)**
1	60	25	0.2567
2	60	29	0.2157

Table 4.2 Impact Test Result

CHAPTER 5

CONCLUSION

The present investigation on LM series aluminium alloy hybrid composites reinforced with zirconium and graphite has demonstrated significant improvements in both mechanical and tribological properties. The stir casting process proved to be an effective and economical method for fabricating the composites with fairly uniform distribution of reinforcements. The addition of zirconium contributed to grain refinement, which enhanced the tensile strength and hardness of the material, while graphite improved wear resistance due to its inherent self-lubricating properties, thereby reducing friction during operation. The combined use of zirconium and graphite exhibited a synergistic effect, resulting in a balanced enhancement of strength, hardness, and wear behavior. However, at higher reinforcement percentages, minor agglomeration and a slight reduction in ductility were observed. Overall, the developed hybrid composite shows strong potential for use in automotive, aerospace, and structural applications where lightweight materials with improved performance are required.

CHAPTER 6
FUTURE SCOPE

- Investigation of different reinforcement percentages to determine the optimal composition for specific applications.
- Study of high-temperature behavior and thermal stability of the composites.
- Use of advanced characterization techniques like SEM, XRD, and TEM for detailed microstructural analysis.
- Exploration of alternative reinforcement materials such as silicon carbide (SiC), alumina (Al_2O_3), or boron carbide (B_4C).
- Analysis of corrosion resistance and fatigue behavior under real working conditions.
- Application of machine learning models to predict material properties based on composition and processing parameters.
- Scaling up the process for industrial manufacturing and evaluating cost-effectiveness.

CHAPTER 7

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